

[PHYS 6268] Nonlinear Dynamics: Problem Set 12

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Strogatz Book: Problems 10.1.11, 10.2.6, 10.3.11, & 10.4.1

1 Problem 10.1.11: Cubic Map

Consider the map

$$x_{n+1} = 3x_n - x_n^3 \quad .$$

1.1 Part (a): Find all fixed points and classify their stability.

Fixed points of the system occur when the mapping returns a value equal to the input. Expressed in terms of the map above, this implies

$$x_n^* = x_{n+1} = f(x_n^*) \quad \implies \quad x_n^* = 3x_n^* - x_n^{*3} \quad .$$

Hence, we have a cubic polynomial expression that admits between one and three roots that each correspond to a fixed point. Rearranging the above, we find

$$x_n^{*3} = 2x_n^* \quad ,$$

which immediately implies that $x^* = 0$ is a fixed point of the map. Then, dividing through by x_n^* (since we have accounted for the possibility it vanishes), we arrive at

$$x_n^{*2} = 2 \quad \implies \quad x^* = \pm\sqrt{2} \quad .$$

Hence, the fixed points of the mapping f are $[0, -\sqrt{2}, \sqrt{2}]$.

In order to classify the equilibria stability, we linearize the map via a Taylor expansion. This yields a linearized mapping $\eta_{n+1} = f'(x^*)\eta_n$. Therefore, this expression definitively retains its stability as the map is iterated in the neighborhood of the fixed point only if $|f'(x^*)| < 1$. In the event that $|f'(x^*)| = 1$, higher order (quadratic) stability analysis is required. For all values of $|f'(x^*)| > 1$, the map is unstable in the neighborhood of the fixed point x^* .

Thus, we calculate $|f'(x_n)|$ and evaluate it at the fixed points:

$$f'(x_n) = 3 - 3x_n^2 \quad .$$

Taking the magnitude of this expression, we find that the *eigenvalue* $\lambda \equiv |f'(x^*)|$ (Strogatz) of the map is given by

$$|f'(x_n)| = |3 - 3x_n^2| \quad .$$

Hence,

$$\begin{aligned} |f'(x^*)| &= |f'(0)| = |3 - 0| = 3 \\ |f'(x^*)| &= |f'(\pm\sqrt{2})| = |3 - 3 \cdot 2| = |-3| = 3 \end{aligned}$$

in both cases, our eigenvalue is greater than unity, and our mapping is not generally stable to small perturbations in the neighborhood of any of the three fixed points.

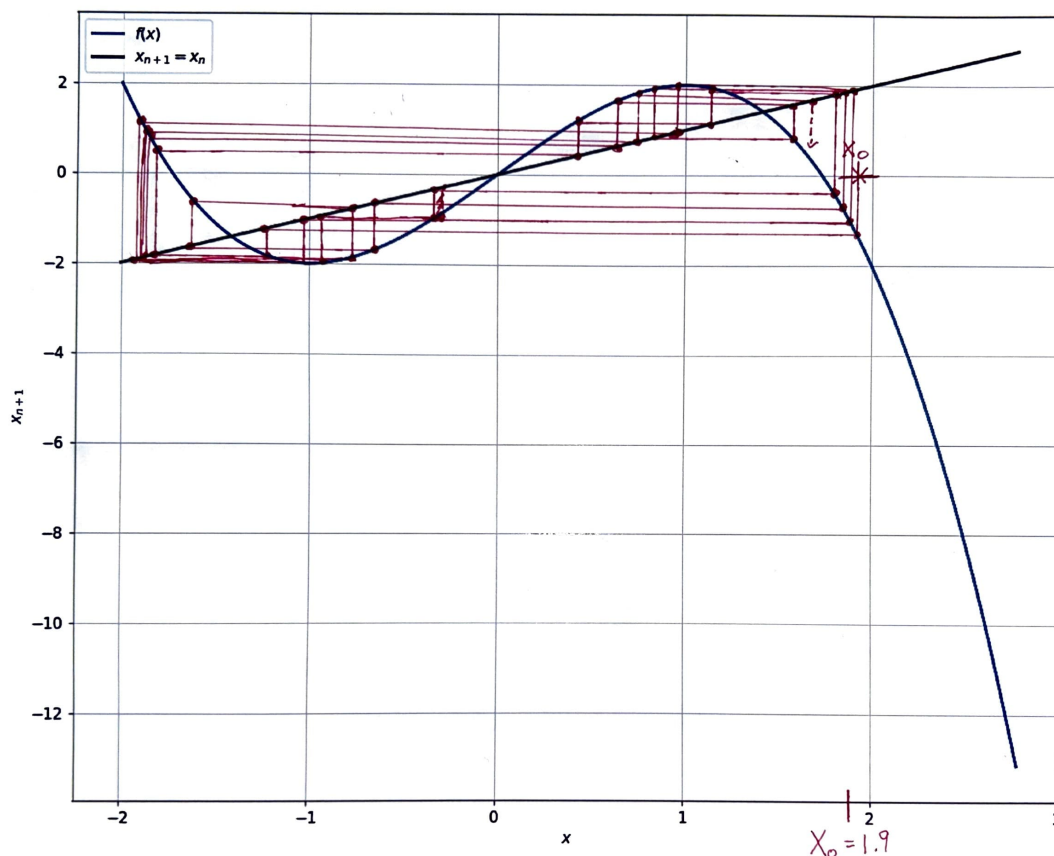


Figure 1: “Cobweb” construction for the mapping $f(x_n) = 3x_n - x_n^3$. As indicated at the bottom of the horizontal axis, the initial condition is $x_0 = 1.9$. Since the initial value of the map is negative, the start of the construction flows “downward”. The tracing procedure is carried out until the density of curves in certain regions makes it difficult to distinguish the trajectory on the Figure. What is shown here is sufficient to elucidate the end behavior of the system for this initial condition.

1.2 Part (b): Draw a cobweb starting at $x_0 = 1.9$

To do so, we generate a plot of our curve $f(x_n)$ superimposed with the line $x_n = x_{n+1}$. Then, we trace the cobweb with a pen, scan the printed plot with the cobweb “annotations”, and include it here as a Figure. The cobweb starting at $x_0 = 1.9$ is included in Figure 1.

1.3 Part (c): Draw a cobweb starting at $x_0 = 2.1$

To do so, we repeat the process described in Section 1.2. The cobweb starting at $x_0 = 2.1$ is included in Figure 2.

1.4 Part (d): Try to explain the dramatic difference between the orbits found in parts (b) and (c).

The primary difference between the two “cobwebs” in Sections 1.2 and 1.3 is the end behavior of the trajectory. As can be seen by comparing Figures 1 and 2, the value of $x_0 = 2$ represents a critical point in parameter space for the initial condition of this map. For values between $x_0 = \sqrt{2}$ and $x_0 = 2$, the trajectory has the “opportunity” to remain bounded in this region of parameter space for a substantial amount of time. This is afforded by the image of the $f(x_n)$ curve reaching down to values as low as -2 in the $x < 0$ half space.

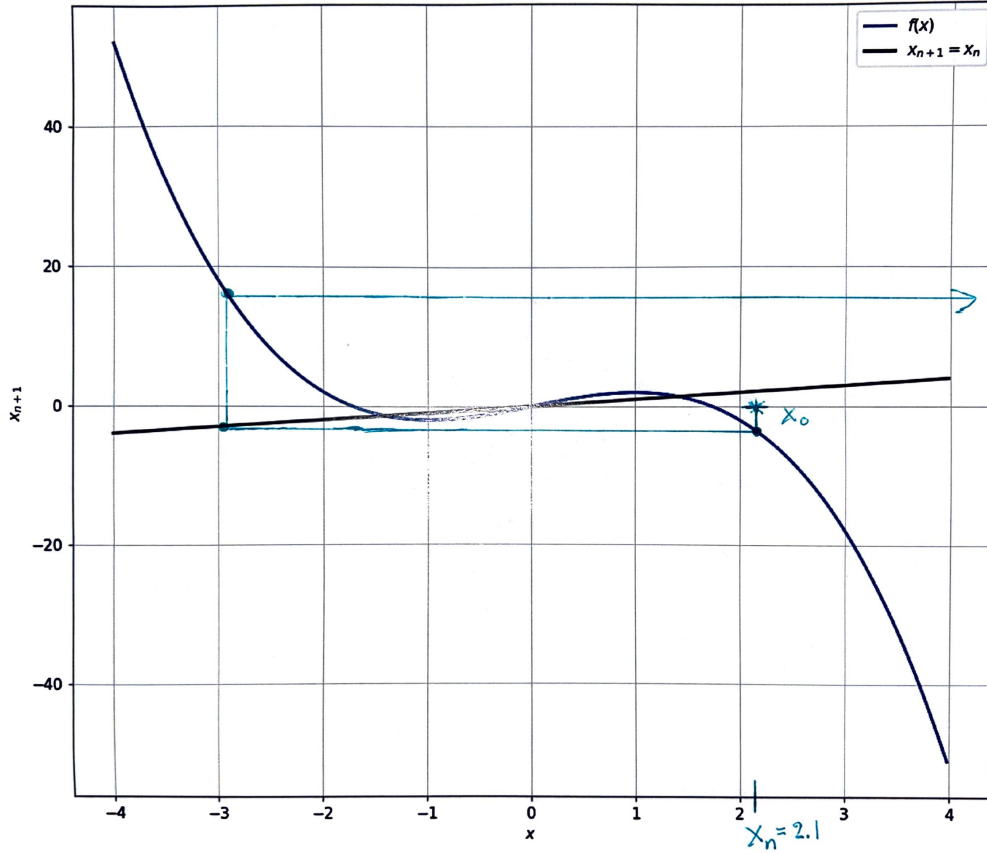


Figure 2: “Cobweb” construction for the mapping $f(x_n) = 3x_n - x_n^3$. As indicated at the bottom of the horizontal axis, the initial condition is $x_0 = 2.1$. Since the initial value of the map is negative, the start of the construction flows “downward”. Due to the unstable nature of this trajectory, we restrict the domain far less in this Figure than in Figure 1. The trajectory continues to exhibit a self-similar pattern across larger and larger spatial scales as it diverges.

In contrast, the trajectory for values of $x_0 > 2$, we can see an entirely different behavior from the first few segments of the cobweb construction. First, we recognize that $|f(x_n)| > x_n$ for all $x_n > 2$. This implies that any input to the map will return an output that is *larger*. Since there are no equilibria or critical points for values of $x_n > 2$, this behavior will continue indefinitely. As can be seen from the cobweb construction in Figure 2, this behavior is reflected by the example.

In order to make our statements above slightly more rigorous, let us verify that the mapping is unstable to arbitrary inputs $|x_i| > 2$. We have that

$$f(x_i) = 3x_i - x_i^3 \quad ,$$

which we can see is dominated by the cubic term above input values on the order of unity. In order to compare the output to the input, we define the function serving as a ratio of the two $R(x_n)$ given by

$$R(x_n) = \frac{3x_n - x_n^3}{x_n} = 3 - x_n^2 \quad .$$

Since we are only interested in general divergence, not the direction, we care about

$$|R(x_n)| = |3 - x_n^2| \quad .$$

This quantity grows unbounded as x_n approaches infinity. This means that successive inputs will be further and further separated, implying the divergent end behavior is only exaggerated as x_n grows.

We can also recognize that for our value $x_i > 2$, we have

$$|R(x_i)| = |3 - x_i^2| > |3 - 4| = 1 \quad ,$$

which implies that the output of the mapping is *always* larger than the input for x_i . Due to the magnitudes of the quantities being the only relevant value, this argument similarly applies for initial points on the map $x_i < -2$. Therefore, combining the two arguments shows that, regardless of the input being positive or negative, as long as $|x_0| > 2$, values of the map will always satisfy $|f(x_n)| > |x_n|$, and thus it is clear that a sequence of values $\{x_0, x_1, x_2, \dots\}$ admits the properties

$$\begin{aligned} |x_{n+1}| &> |x_n| \\ ||x_{n+2}| - |x_{n+1}|| &> ||x_{n+1}| - |x_n|| \end{aligned}$$

which indicate that

$$x_n \longrightarrow \infty \quad \text{if} \quad |x_0| > 2 \quad ,$$

which immediately applies to the map trajectory in Section 1.3 and Figure 2.

Because this argument *does not apply* to the map trajectory in Section 1.2, we would not expect the same behavior there, consistent with what we show in Figure 1.

2 Problem 10.2.6: Period-doubling and Chaos Galore

Consider the map

$$x_{n+1} = r \cos x_n \quad .$$

Plot the orbit diagram for this map. Be sure to use a large enough range for both r and x to include the main features of interest. Also, try different initial conditions, just in case it matters.

The orbit diagram is included in Figure 3.

3 Problem 10.3.11: Subcritical Flip

Let

$$x_{n+1} = f(x_n), \quad \text{where} \quad f(x_n) = -(1+r)x_n - x_n^2 - 2x_n^3 \quad .$$

3.1 Part (a): Classify the linear stability of the fixed point $x^* = 0$.

To do so, we follow the same linearization procedure for the mapping f as we described in Section 1.1. We have that

$$f(x_n) = -(1+r)x_n - x_n^2 - 2x_n^3 \quad ,$$

which implies that

$$f'(x_n) = -(1+r) - 2x_n - 6x_n^2 \quad .$$

Since the absolute value of f' is what determines convergence, we write

$$|f'(x_n)| = |-(1+r) - 2x_n - 6x_n^2| = |-(1+r) + 2x_n + 6x_n^2| = |(1+r) + 2x_n + 6x_n^2| \quad .$$

Evaluating $|f'(x_n)|$ at the fixed point, we find

$$|f'(x^* = 0)| = |(1+r)| \quad .$$

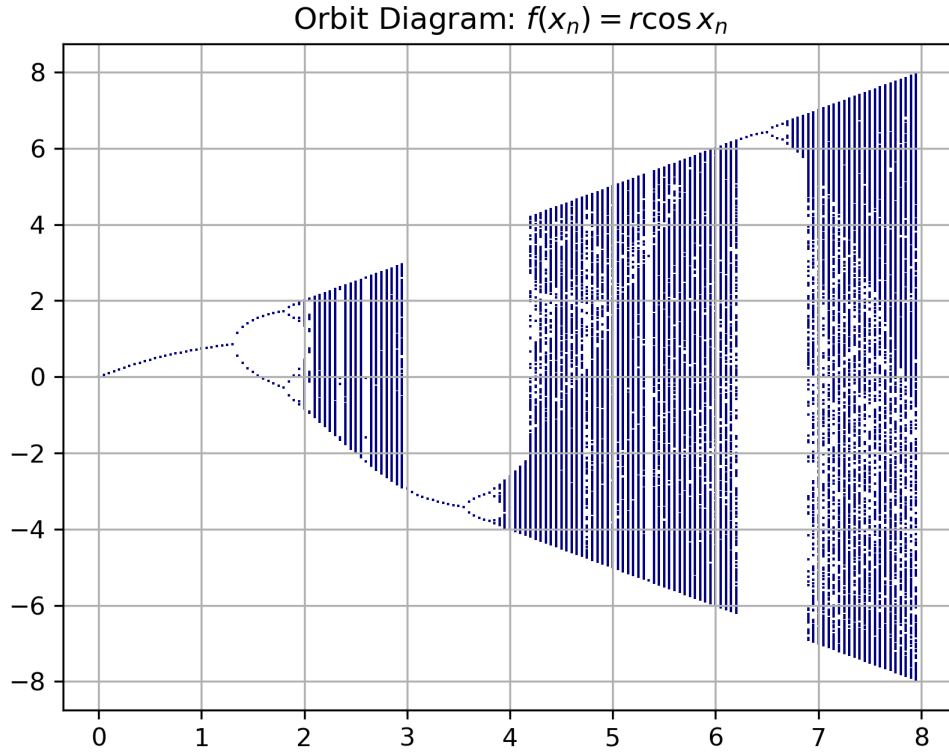


Figure 3: Orbit diagram for $f(x_n) = r \cos x_n$. The period-doubling phenomenon is visible along the branches. Between them, there are periodic windows that appear as gaps in points.

Therefore, we can constrain the stability of the mapping f as a function of the parameter r . We see that

$$\begin{aligned} \text{if } r \in (-2, 0) &\longrightarrow |f'(x^*)| < 1 &\implies &\text{stable} \\ \text{if } r \in (-\infty, -2) \cup (0, \infty) &\longrightarrow |f'(x^*)| > 1 &\implies &\text{stable.} \end{aligned}$$

Hence, the map is stable in the neighborhood of the fixed point $x^* = 0$ if $r \in (-2, 0)$.

For the borderline values $r = -2$ and $r = 0$ where $|f'(x^* = 0)| = 1$, we use $|f''(x^* = 0)|$ to determine that $r = 0$ is *unstable* and that $r = -2$ is also *unstable*. In fact, the second order stability (concavity) does not even depend on r since the term dependent on r is only *linear* in x_n .

3.2 Part (b): Show that a flip bifurcation occurs at $x^* = 0$ when $r = 0$.

Since as $r \rightarrow 0^-$ (from the left), $|f'(x^*)|$ passes through unity from lower to higher values, the stability of x^* is sensitive to the point $r = 0$. The behavior near the fixed point changes from stable to unstable (as indicated in Section 3.1) when r passes through zero from below. Hence, a bifurcation around the equilibrium $x^* = 0$ occurs at $r = 0$.

A flip bifurcation occurs when a parameter causes the slope $f'(x^*)$ to pass through the value of -1 at the intersection with the curve $x_{n+1} = x_n$. At the point where this occurs, stable, damped oscillations toward the fixed point grow unstable, typically involving period doubling phenomena.

Hence, to assert that the bifurcation is a flip bifurcation, we calculate

$$\begin{aligned} f'(x_n) &= -(1+r) - 2x_n - 6x_n^2 \\ \implies f'(x^* = 0) &= -(1+r) \\ \implies f'(x^* = 0)|_{r \rightarrow 0} &= -1 \quad , \end{aligned}$$

where we have verified that the linear behavior of the map matches our expectations. For $r < 0$, we have that $f'(x^* = 0) > -1$, and as $r \rightarrow 0$ from negative values, we see that $f'(x^* = 0) \rightarrow (-1)$. Hence, a flip bifurcation does indeed occur at $x^* = 0$ when $r = 0$.

3.3 Part (c): Show that there is an *unstable* 2-cycle for $r < 0$, and that this cycle coalesces with $x^* = 0$ as $r \rightarrow 0^-$ (from below).

Via Strogatz, a 2-cycle exists *if and only if* $\exists p, q \in \mathbb{R}$ such that

$$f(p) = q \quad \text{and} \quad f(q) = p \quad .$$

Equivalently, any such point p must simultaneously satisfy

$$f(f(p)) = p \quad .$$

That is, p must be a fixed point of the *second-iterate* map $f^2(x_n) \equiv f(f(x_n))$.

Then, solutions p, q are points where $f^2(x) = x$. The fixed points of the original map f are given by

$$x^* = 0, -\frac{1}{4} \pm \frac{\sqrt{-8r-7}}{4}$$

and can be factored out of our second-iterate map, since they must be fixed points of f^2 if they are fixed points of f .

In our case, the second-iterate map f^2 is given by

$$\begin{aligned} f^2(x_n) &= f(f(x_n)) = f(-(1+r)x_n - x_n^2 - 2x_n^3) \\ &= (1+r)((1+r)x_n + x_n^2 + 2x_n^3) - (-(1+r)x_n - x_n^2 - 2x_n^3)^2 + 2((1+r)x_n + x_n^2 + 2x_n^3)^3 \quad . \end{aligned}$$

which admits five unique roots, each corresponding to possible p, q values. However, we require that p, q are not equilibria for the original system, or else, this would not be a 2-cycle. Hence, we discard the roots $x^* = 0, -\frac{1}{4} \pm \frac{\sqrt{-8r-7}}{4}$, leaving two distinct roots corresponding to p and q in terms of r . The conjugate pair is a set of distinct, real roots if $r < 0$, but the roots coincide to become complex conjugates the moment that $r \rightarrow 0^-$. The real part approaches the origin as the p and q branches of the cycle coalesce with x^* simultaneously, indicating they bifurcate into the fixed point $x^* = 0$ (which exists $\forall r \in \mathbb{R}$).

To demonstrate that the 2-cycle is not stable, we must evaluate the linearization of the second-iterate map $|f'(q)f'(p)|$ as a function of r :

$$|f'(q)f'(p)| = |((1+r) + 2p + 6p^2)((1+r) + 2q + 6q^2)| \geq \left| (1+r)^2 + \frac{1}{2} + \frac{3}{8} + \frac{1}{8} \right| \geq |(1+r)^2 + 1|$$

using the fact that p, q are conjugates. Evaluating the above, we find that $|f'(q)f'(p)| > 1$ when $r < 0$, indicating that our 2-cycle is unstable.

3.4 Part (d): What is the long-term behavior of orbits that start near $x^* = 0$, both for $r < 0$ and $r > 0$.

The long term behavior near the fixed point x^* is sensitive to the sign of r , but *not* the sign of x_0 . The successive iterations of the mapping f , in both cases (positive vs. negative r), will oscillate

between positive and negative values of x_{n+1} . The distinction, however, comes in the absolute value of the oscillations: for a value of $r < 0$, the magnitude of successive steps satisfies

$$|x_{n+1}| < |x_n| \quad .$$

In this case, the fixed point x^* is stable, and the system will settle down to the equilibrium from the nearby neighborhood facilitated by dampened oscillations.

Alternatively, for $r > 0$, successive iterations of the mapping satisfy

$$|x_{n+1}| > |x_n| \quad ,$$

because the fixed point x^* now is unstable in this region of parameter space. Therefore, trajectories will be forced away through oscillations about $x^* = 0$ with a successively growing amplitude.

4 Problem 10.4.1: Exponential Map

Consider the map

$$x_{n+1} = re^{x_n} \quad ,$$

for $r > 0$.

4.1 Part (a): Analyze the map by drawing a cobweb.

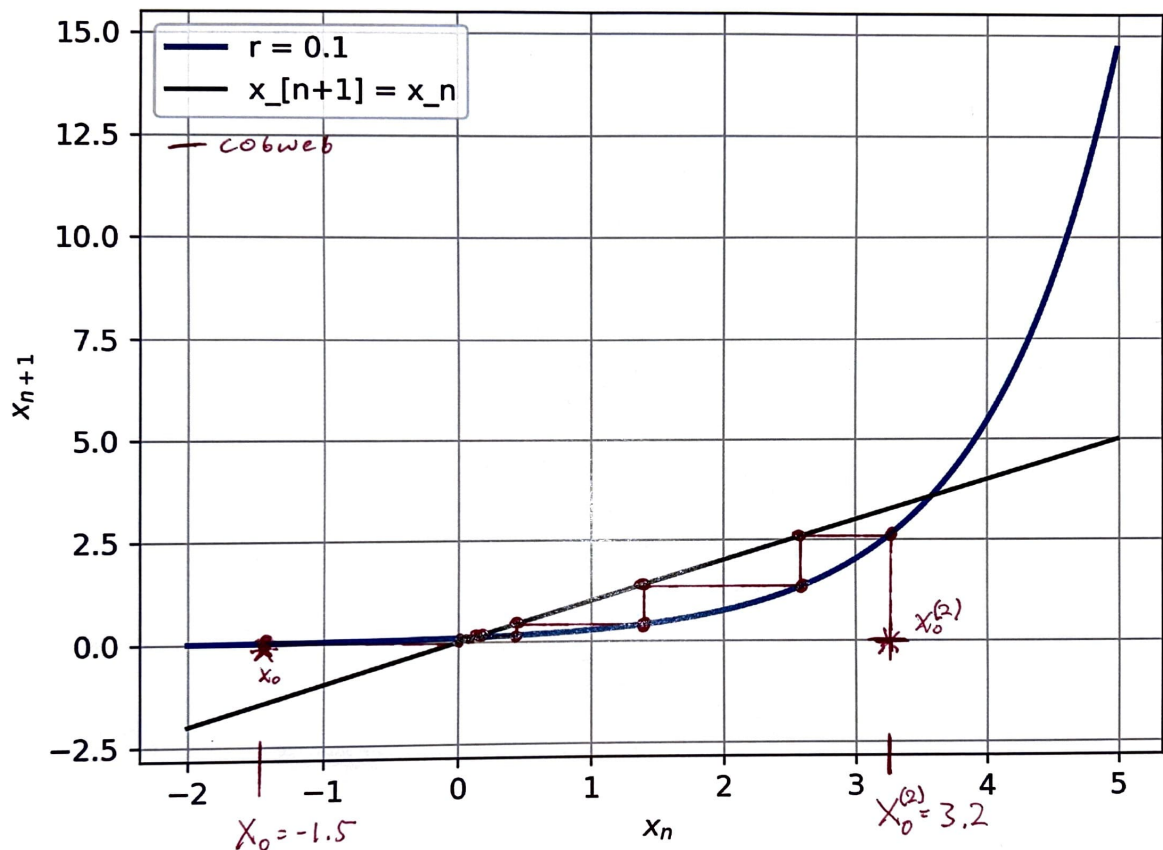


Figure 4: “Cobweb” construction for the mapping $f(x_n) = re^{x_n}$, using a value of $r = 1/10$. As indicated at the bottom of the horizontal axis, the initial condition is $x_0 = -1.5$ for the first construction, and $x_0^{(2)} = 3.2$.

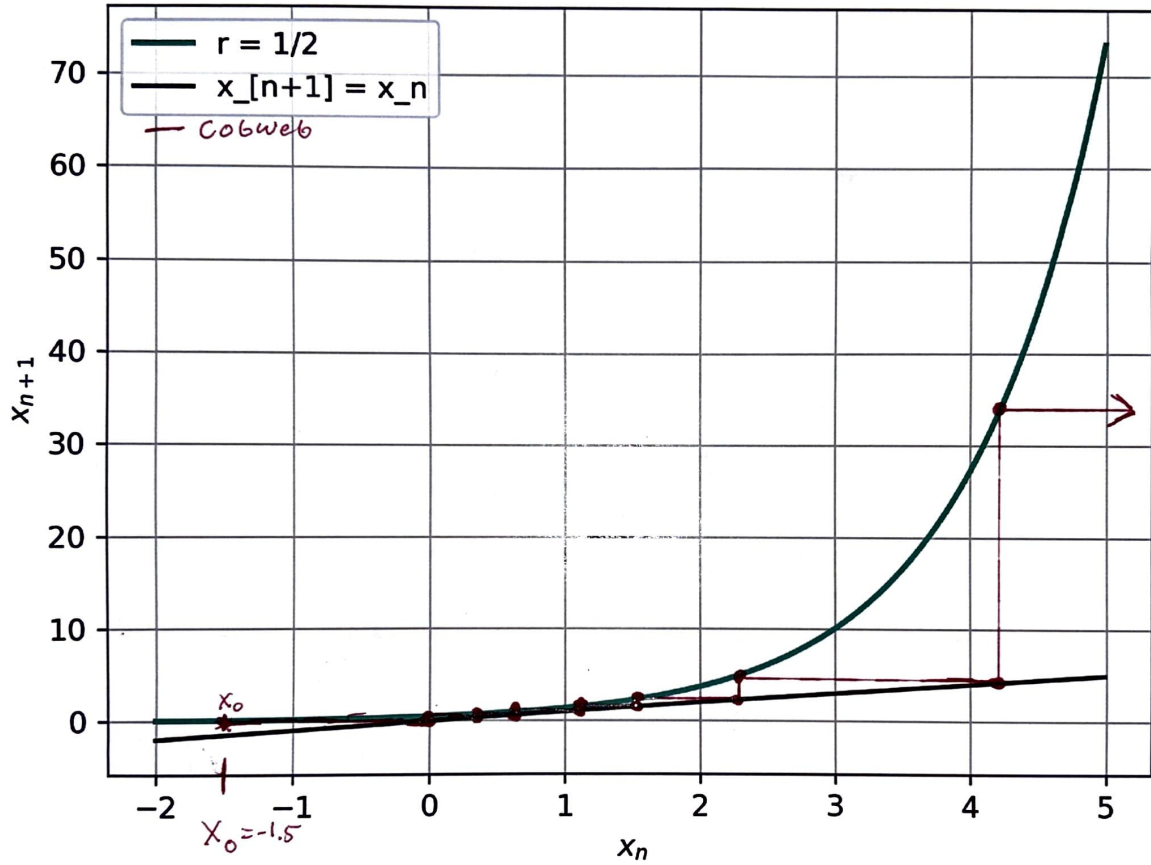


Figure 5: “Cobweb” construction for the mapping $f(x_n) = re^{x_n}$, for $r = 1/2$. As shown by the horizontal axis, the initial condition is $x_0 = -1.5$.

To do so, we adopt a procedure akin to that employed in sections 1.2 and 1.3. We select two values of r for which to construct a cobweb. The cobweb for $r = 1/10$ is depicted in Figure 4. The cobweb for $r = 1/2$ is shown in Figure 5.

4.2 Part (b): Show that a tangent bifurcation occurs at $r = 1/e$.

As can be seen in the cobweb constructions from section 4.1, a bifurcation of the system must occur somewhere between $r = 1/10$ and $r = 1/2$, since the system behavior completely changes. For $r = 1/2$, the curve $f(x_n)$ is always larger than x_n . As can be seen in Figure 5, the cobweb construction spends many iterations in the neighborhood around $x_n \in (0, 1)$ where the output of the map is only *slightly* larger than the input. However, eventually the trajectory “breaks out” of this region and the successive inputs grow at an increasing rate. This is exemplified by the last few segments of the trajectory that are growing progressively longer.

On the other hand, Figure 4 clearly illustrates an entirely distinct behavior for the system (from the same initial condition) when $r = 1/10$. In this case, there is a region of small, positive x_n where $x_n > f(x_n)$. This region is critical to allow the trajectories in the cobweb to converge towards the intersection of $f(x_n)$ and x_n . Both constructions in this figure, starting from opposite sides of $x_0 = 0$, eventually converge towards this point. Moreover, by inspection and comparison of the two curves for f in Figures 4 and 5, it is highly plausible that for some critical value $r_c \in [\frac{1}{10}, \frac{1}{2}]$, a tangent bifurcation occurs in this region.

A tangent bifurcation is characterized by the coalescence of fixed points or cycles into a single point that vanishes at the moment the map satisfies $f'(x^*) = 1$. At the fixed point x^* , the instant of the bifurcation involves the mapping $f(x)$ being tangent to the curve $x_{n+1} = x_n$, motivating the bifurcation's name. Therefore, we seek to show that $f'(x^*) = 1$. The derivative of the map is given by

$$\frac{df}{dx} = re^x \quad ,$$

which equals unity when

$$x = \log\left(\frac{1}{r}\right) \quad .$$

However, for it to be a fixed point, we also require that

$$x = re^x \quad \implies \quad r = xe^{-x} \quad .$$

Therefore, we can substitute

$$x = \log\left(\frac{1}{xe^{-x}}\right) = \log\left(\frac{e^x}{x}\right) \quad .$$

This implies

$$x = x - \log x \quad .$$

Thus, the fixed point x^* is located at $x^* = 1$. The critical value of the parameter can be readily calculated:

$$r_c = x^*e^{-x^*} = \frac{1}{e} \quad .$$

Therefore, a tangent bifurcation occurs for the map f at the critical value of $r_c = 1/e$. The tangent bifurcation occurs at the fixed point $x^* = 1$.

4.3 Part (c): Sketch the time series x_n vs. n for r just above and just below $r = 1/e$.

The timeseries is included in Figure 6.

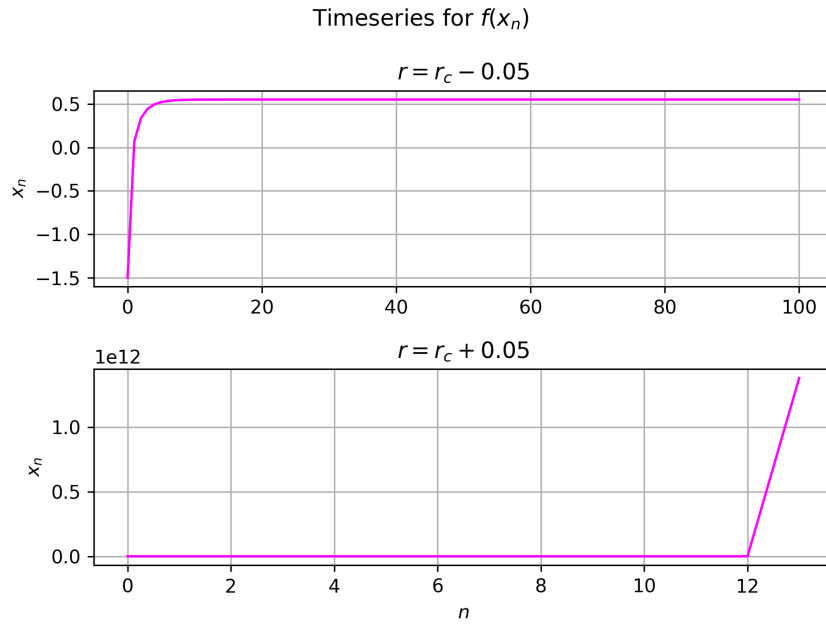


Figure 6: Timeseries for the map $f_r(x_n)$ for values of $r = r_c - \epsilon$ (top) and $r = r_c + \epsilon$ (bottom). As can be seen in the Figure, the subcritical value causes the system to tend towards a constant value around 0.6, whereas the supercritical value of r sees the mapping diverge (overflow in python float type) after only 12 iterations.

5 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the integration of the Lorenz system (see Section 2), as well as plotting: these were completed using python (`numpy` and `matplotlib`) libraries.